

Private Standards and Labour Productivity in the Food Sector in Vietnam

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1ST DRAFT, MARCH 2016

Abstract

A rising number of firms from developing countries have adopted voluntary private standards in the last decade. This has become an area of active research, especially in terms of the impact of private standards on trade, organisational performance and employee outcomes. This paper analyses how standards affect labour productivity on a three-year panel data of small and medium firms from the food sector in Vietnam. To identify the causal effect of standards, the estimation uses fixed effects and a difference GMM approach. The results show that the application of private standards improves labour productivity. These gains primarily occur to firms operating above a certain labour-intensity level. Firms with low labour intensity are not likely to experience gains in labour productivity from standards. This implies that employee compensation increase due to standards is a likely mechanism for labour productivity gains.

Keywords: standards, labour productivity, food, SMEs, Vietnam

JEL codes: L66, D22, J4, J81, O12

1. Introduction

Food production and trade have become inseparable from requirements for certification of standards that regulate quality, safety, social or environmental impact of products and production processes. Standards have emerged as a way of improving consumer's information about product characteristics, affecting consumer loyalty and trust (Raynolds, 2002). This is especially relevant for food trade from developing to developed countries (Beghin et al., 2015). Private standards are applied voluntarily and, per occasion, independently from national regulation. The issues they address may or may not overlap with the official regulation. Standards are found to substitute for missing public regulation, a process described as 'a shift from public to private governance' (Hatanaka et al., 2005).

Firms in the food sector put a lot of effort in assuring compliance with private standards, which can potentially improve access to higher-value markets (Masakure et al., 2009), firm's reputation (Fulponi, 2006), financial performance (Alpay et al., 2002; Corbett et al., 2005; Foster and Gutierrez, 2013) and competitiveness (Delmas, 2001), but only if firms can overcome the costs of implementation (Maskus et al., 2013). The inability to finance compliance with standards has been identified as one of the main obstacles for participation of small-scale producers from developing countries in global trade (Henson and Humphrey, 2010).

A growing body of literature is focusing on the impact of standards on firm performance, covering both developed and developing countries. Fontagné et al. (2015) analysed the impact of standards on export performance of French firms, while Martincus et al. (2010) and Otsuki (2011) investigated the effect ISO certification on export performance of firms in Argentina and in Europe and Central Asia. Schuster and Maertens (2015) analysed the effect of various types of private standards on export performance of firms in Peru using fixed effects and GMM models. Henson et al. (2011) and Masakure et al. (2009) analysed the returns to certification in terms of export sales revenue for sub-Saharan African countries and Pakistan. Apart from the study in Peru, all studies report positive effects of standards on export performance and revenue. Goedhuys and Sleuwaegen (2013) studied how international standards certification affects productivity and sales performance in various countries. They found that certification raises productivity and sales, with the effects being larger in countries where market supporting institutions are weak.

The growth of high-standards food exports from developing countries has been associated with positive welfare outcomes and extended employment opportunities (Beghin et al., 2015; Maertens and Swinnen, 2009). Especially in high-value export sectors, private standards can lead to better employment conditions.

Colen et al. (2012) have associated GlobalGAP certification with higher employee daily wages and longer employment periods in exporter–producer companies in Senegal. Blunch and Castro (2005) have found that ISO 9000 and ISO 14000 certification affects firm’s training decisions. Schuster and Maertens (2016) show mixed evidence of labour standards in Peru, such as Ethical Trading Initiative (ETI) and Social Accountability 8000 (SA8000): While food export firms with labour standards appear more likely to pay minimum wage, they are not likely to offer higher wages or longer employment. How these benefits relate to labour productivity is fairly underexplored in the existing literature.

Looking at labour productivity enables understanding whether private standards introduce differences in the efficiency of labour input use between certified and non-certified firms. Labour productivity can change due to the adjustments in the firm's work system after the implementation of standards. For example, international standard ISO 9001 optimises processes while ISO 22000 can cut operational costs by managing food safety risks. If standards entail employee skill-building and streamline operating procedures, cut waste and increase sales, the changes will be registered as improved labour productivity. Opposite, if the costs of implementation of standards are too high, additional employee effort and skills may be under-rewarded and stall productivity. In competitive markets, the differences in productivity resulting from the investments in standards would be entirely reflected in wage differentials. In practice, however, the relationship between gains in productivity and wages can vary according to the origin of financing, job type, as well as wage and fringe benefits structure. In the case of standards, it is probable that there is a considerable divergence between wages and productivity gains since it is the employers who bear the costs of standards implementation. Thus, the wage premium attributed to standards in earlier studies is likely to constitute a lower bound of productivity gains resulting from this investment.

I look at the effect of standards on labour productivity, where labour productivity is measured as value added per worker. The paper uses a panel dataset from three rounds of SME surveys in Vietnam, conducted in 2011, 2013 and 2015, covering 2,144 observations (988 firms) in the unbalanced and 1,425 observations (475 firms) in the balanced panel. To identify the causal effect of standards, the estimation employs fixed effects and a difference GMM approach. The results show that the application of private standards improves labour productivity among the SMEs from the food sector in Vietnam. Firms that have adopted private standards enjoy 20-37% higher labour productivity than firms that have not adopted such standards. The paper also suggests that the benefits from standards are higher for firms with higher labour compensation, implying that employee wage increase due to standards is a likely mechanism for labour productivity gains. The results supplement earlier findings of a positive impact of standards on employee

outcomes (Colen et al., 2012; Delmas and Pekovic, 2013; Levine and Toffel, 2010; Schuster and Maertens, 2016). Slowing income and economic growth in many developing countries have been attributed to the inefficient labour use and a lack of productivity growth (Rodrik, 2011). By linking standards and labour productivity, this paper brings a policy-relevant perspective on the performance of the SME sector, which, as Beck et al. (2005) argue, is the foundation of the employment and economic growth for developing countries.

2. The Vietnamese food sector

Food processing is one of the most important manufacturing sectors in Vietnam as it employs around 10% of all workforce in manufacturing. The sector has grown four times in value at current prices since 2005, as illustrated in Figure 1. The growth has been around 5% in recent years: the industry has expanded by 5.1% in 2014 and by 6% in 2013 (at 2010 prices). The number of firms in the food sector was around 5,000 in 2005 and it has increased to 5,820 in 2013, showing a 16% overall increase or annual growth of 3.1%.

[Figure 1 around here]

The data from Vietnam's General Statistics Office (GSO) show that the food types processed in largest volumes are: milled rice (45 million tons), refined sugar (1.8 million tons) and frozen aquatic products (1.6 million tons) (GSO, 2015a). Processing increased sharply between 2005 and 2014: production of refined sugar increased by 70%, milled rice by 50% and frozen aquatic products by 132%. The highest annual growth rate of 13% was observed for fresh milk, produced at 840 million litres.

[Figure 2 around here]

Food sector contributed to about 15% of the total export value in 2013. Figure 3 shows that the growth in food manufacturing was followed by a considerable growth in the export of food, agricultural and aquatic products. Food exports reached 18.6 billion USD in 2013. As a comparison, all exported manufactured products were valued at 98.1 billion USD in 2013, with textiles and garment taking 20.8 billion USD and footwear exports taking 10.2 billion USD (GSO, 2015a). The value of food export has grown nine times between 1995 and 2013, with yearly expansion of 12%. The US and the EU are Vietnam's two largest export markets.

[Figure 3 around here]

3. Conceptual background

Certification of private standards facilitates activities between different economic agents (e.g. suppliers and buyers) by transferring non-market information about product and process characteristics, such as product quality and safety or the way a firm treats environment and labour. This paper considers private standards set by non-governmental bodies that are verified through third-party certification in which a certificate is awarded after a production process is checked by authorized independent domestic or international persons or organisations. Internationally recognised standards reduce cross-country differences of national government regulations, so they are assessed as a useful governance mechanism for firm behaviour (Christmann and Taylor, 2006). Private standards usually require that firms implement specific activities that go beyond what is required by the national regulatory framework. Implementation of ISO standards usually includes: examining adequacy of work processes and methods for meeting product specifications; documenting work processes, work instructions and quality assurance procedures; internal auditing to verify that activities comply with the procedures; and designing preventive and applying corrective actions in response to audits (Naveh and Erez, 2004).

Labour productivity can change due to the adjustments in the firm's work system after implementation of standards. It is not a priori known how standards will affect labour productivity. The implementation of standards requires organisational changes, improving operating procedures, management practices and production processes, which translate into increased sales (Levine and Toffel, 2010). It was shown in other contexts that organisational changes lead to better productivity (Caliendo and Rossi-Hansberg, 2012). Naveh and Marcus (2005) reported that standards lead to improved operating performance measured by lower defect rates, reduced cost of quality, on-time delivery and customer satisfaction. Standards lower transaction costs between trade partners in global value chains, which helps to increase customer demand. They also enable firms to charge a price premium for investments in the innovations put in place in order to obtain certificates (Paunov, 2016). Furthermore, firms are required to train employees about quality, safety, record-keeping, implementation of new procedures and audits. These potentially manifest in higher levels of interpersonal interaction or greater employee engagement in business operations, which can lead to increased job satisfaction, more effective knowledge transfer or innovative ideas that improve productivity (Delmas and Pekovic, 2013). Indeed, Elmuti and Kathawala (1997) found support for

improvements in quality of work life and several studies found wage increase due to certification (Colen et al., 2012; Levine and Toffel, 2010; Schuster and Maertens, 2016). However, wage and productivity increase due to standards need not follow parallel trends.

If the costs of implementation of standards are too high, compensation for additional employee effort and skills may be insufficient to trigger positive productivity effects. Antle (2000) showed that food safety regulations increased production costs in the US meat industry. The costs increased because of the necessary modifications of the production process and a loss in operating efficiency. Similar findings are evident in other US industries, such as poultry (Goodwin Jr. and Shiptsova, 2002). Other costs associated with standards include repairs, documentation, audits and certification. Such costs are expected to vary with product and firm size, as well as the sector and geographical location (Masakure et al., 2009). It is usually not enough to comply with one standard, but producers are required to certify several standards, each likely targeted at particular destination market, which can further increase costs. The costs related to standards are especially burdensome for small firms (Wang et al., 2009). It is also argued that the gains from standards come at the expense of workers' earnings. Formalisation and documentation of work practices that accompany implementation of standards can negatively affect workers by creating routinized work places, with lower skill requirements, which are making individual workers more easily replaceable and decrease worker bargaining power (Brenner et al., 2004; Levine and Toffel, 2010).

The empirical evidence on the impact of private standards and productivity is ambiguous. Environmental and food safety regulation are argued to have a negative impact on firms' performance (Bontemps et al., 2012), but it may not always be so. Several studies attribute productivity improvements to standards and certification (Alpay et al., 2002; Corbett et al., 2005; Elmuti and Kathawala, 1997; Foster and Gutierrez, 2013; Lanoie et al., 2008; Porter and van der Linde, 1995). Considering that standards introduce changes in firm operations and processes, they can be considered as innovation, but just as with other types of innovation, the effect could go in any direction. Indeed, the evidence of the impact of innovation on productivity is mixed. In a study of Spanish manufacturing firms, Huergo and Jaumandreu (2004), as well as Cassiman et al. (2010) found that process innovation caused productivity increase, while Cozzarin (2016) found a negative effect of product innovation on productivity among Canadian firms. Van Leeuwen and Klomp (2006) found that process innovation does not increase labour productivity.

4. Data

The analysis is based on the data from the small- and medium-sized enterprise (SME) survey from Vietnam. This survey has been conducted every second year since 2005 with the aim of evaluating characteristics of the Vietnamese business environment. It is implemented in 10 provinces in Vietnam: Ho Chi Minh City (HCMC), Hanoi, Hai Phong, Long An, Ha Tay, Quang Nam, Phu Tho, Nghe An, Khanh Hoa and Lam Dong. The sampling frame comprises a consolidated list of formal enterprises obtained from the Establishment Census from 2002 (GSO, 2004) and the Industrial Survey 2004-2006 (GSO, 2007). Firms are randomly drawn from this list, accounting for ownership type to obtain representative data on household-owned, private, cooperative, limited liability and joint stock enterprises. Apart from the officially registered firms, the survey also includes informal firms that were identified randomly on-site.¹ As the survey traces the same firms over the years, it is able to capture legal structure changes and formalisation of unregistered businesses. Firms who stop operating are randomly replaced based on: (i) the need to maintain a constant level of household firms based on the information in GSO (2004) and (ii) the new 2014 population of firms registered under the Law on Enterprises obtained from GSO (2015b).

The analysis in this paper is based on the data from 2011, 2013 and 2015 survey rounds because the question about the compliance with internationally recognised standards was introduced in 2011. The survey targets non-state manufacturing enterprises from different sectors but I focus on firms registered within manufacture of food products due to disproportionately higher relevance of standards for this sector.² The sample used in the analysis comprises 777 firms in 2015, 714 firms in 2013 and 689 food firms in 2011. Compared to the enterprise census, this sample represents around 12% of all firms in the food sector.³

Informal firms are unlikely to obtain certificate of compliance with standards as certificates normally carry the firm registration number. Indeed, none of the certified firms from the sample are informal. Informal firms, however, make around one-third of the sample and removing these firms from the sample is used as a robustness check of the results.

The main questionnaire includes information on enterprise characteristics and practices. It has stayed almost the same over the years. One notable exception is that the questionnaire 2015 asks about

¹ Detailed information about sampling is available in CIEM et al. (CIEM et al., 2014; 2012; UNU-WIDER, 2015).

² On the importance of standards for food sector, see Lee et al. (2010), Henson and Humphrey (2010) and Beghin et al. (2015).

³ GSO (2014) reports that there were 5,498 registered food firms in 2011 and 5,820 in 2013.

international and domestic standards specifically. All questions refer to the situation in the previous calendar year, namely 2010, 2012 and 2014, while the economic accounts contain information on two consecutive years prior to the survey. The 2011 and 2013 survey rounds only contain an indicator for whether firms apply any of the internationally recognised standards, while the 2015 round reveals which standards exactly are applied. The most frequent are ISO 9001, ISO 14001, HACCP and ISO 22000.

5. Empirical specification

The main goal is to estimate the causal effect of international standards on labour productivity over the period 2010–14. This is done by estimating equation (1):

$$y_{it} = \alpha_i + \beta_i S_{it} + \delta X_{it} + \rho_i + \tau_t + e_{ijt} \quad (1)$$

where i denotes firm and t denotes time period. α_i , ρ_i and τ_t are, respectively, firm, location and time fixed effects. e_{ijt} is the statistical noise term. The dependent variable, y_{it} is the firm-level labour productivity measured as real value added per employee, expressed in 2010 VND. Value added is measured as revenue from sales minus total costs that include expenses on intermediate goods and raw materials and indirect costs. Table 1 shows that the average real value added per employee has increased from around 20 million VND in 2010 to 61 million VND in 2014, achieving an annual growth rate of 25%.

The variable of interest, S_{it} , takes value 1 if a firm applies any private standard and 0 otherwise. The proportion of firms with internationally recognised private standards in the sample is about 5%. The number of certified firms decreased from 2010 to 2014 by around 1.7 percentage points. The most commonly applied standards among the Vietnamese SMEs from the food sector are ISO 9001 and ISO 22000. Only 13 firms (14%) have certified more than one standard.

F_{it} are time-varying firm-specific control variables, such as firm size, value of physical assets and the age of firm. Firm size is controlled for as larger firms have an advantage in complying with standards. A positive size effect on the adoption of standards was found in previous studies (Herath et al., 2007; Masakure et al., 2011; Nakamura et al., 2001). One explanation could be that fixed costs that are bound to be incurred in relation to implementation of standards are less significant for larger firms. Also, due to the well-established labour productivity-size relationship (Van Biesebroeck, 2005), firm size is expected to positively affect labour productivity. Firm size is measured as the total number of regular full-time employees. Summary statistics in Table 1 show that the average firm from the sample employed eight employees and

that the average size has slightly declined between 2010 and 2014. This is in line with the general trend of declining firm size in Vietnam (CIEM et al., 2014).

[Table 1 around here]

Value of capital is also included in the estimation, as a way of controlling for the cost and the nature of technology. This is measured as the deflated value of the total assets of the firm at the end of the year. Table 1 shows that the real value of assets in the surveyed SMEs went up between 2010 and 2014. Firm age is also added as productivity may differ between old and young firms. New firms are shown to have lower average productivity than incumbents (Aw et al., 2001). The average age of firms in the sample is 19 years. The adoption of standards may be influenced by the position in the supply chain, so it is important to control for the type of output, that is, whether a firm produces final or intermediate goods.

Linkages with foreign markets enter estimation as firms are more likely to implement standards if their business is export-oriented. It is also well-established in the literature that export firms have higher productivity (Bernard et al., 2012). Certification improves market access and reduces relational uncertainty and trade costs, enabling creation of long-term trading relationships, which reflect positively on firm performance. The indicator variable takes value 1 if a firm exports any share of its output and 0 otherwise. Table 1 shows that only 3% of the firms in the sample export and that the trend was negative in the 2010-2014 period.

The estimation also controls for legal ownership form as potential benefits can be accrued by changing legal ownership status. Most firms are registered as household establishments (around 80%) and, as most of them are not formally registered, they experience different circumstances related to certification. Legal ownership form enters estimation as a set of dummy variables that represent the specific legal form of the firm (household, private, collective/partnership, limited-liability, or joint-stock enterprise). Table 1 shows small change between 2010 and 2014, the most notable being a decline in the proportion of firms registered as private or sole proprietor establishments and a small rise in all other categories.

Identifying the causal effect of standards on labour productivity requires accounting for the fact that the application of standards is not random among the firms from the sample. The estimation needs to account for simultaneity, whereby firms with already higher levels of labour productivity act more likely to adopt standards. Another difficulty in estimating the causal impact is the presence of unobserved firm-specific

characteristics that influence labour productivity and correlate with the firm's decision to adopt of standards. For example, a manager of a firm may have access to specific information, which could both lead to certification of standards and higher labour productivity.

A typical way of dealing with such issues is fixed effects estimation, which allows controlling directly for all time-invariant unobserved firm-specific factors, such as manager characteristics. Location fixed effects, ρ_p , control for policy changes that may differentially impact productivity of firms in different regions. The estimation controls for the province in which the firm is located through dummy variables, using Ho Chi Minh City as a baseline. This is important because of relative autonomy of Vietnamese provinces, which differ in the extent to which they implement government initiatives (Nguyen et al., 2007). Time dummies, τ_t , control for general trends that affect all firms.

Firms can also have unobservable characteristics, which change over time and which are correlated with both implementation of standards and labour productivity. For example, there may be omitted time-varying firm-specific factors that impact both the decision to implement standards and labour productivity such as, for example, a change in management. In the presence of these factors, standard OLS fixed effects estimates will be biased, but the direction of the bias is not easy to forecast. For example, a change in management could lead a firm to both be more productive and implement standards, in which case OLS estimates will have a positive bias. Alternatively, new management could reduce the extent of activities related to standards in order to invest in productivity-enhancing activities. Also, a demand for certification of private standards coming from trade partners could divert from productivity-enhancing firm decisions. These would lead to a negative bias in OLS estimates.

In addition to a traditional fixed effects approach, I therefore also apply the two-step difference generalized method of moments (GMM) estimator (Arellano and Bond, 1991; Arellano and Bover, 1995; Blundell and Bond, 1998). This method uses first difference transformation to eliminate firm-specific fixed effect (as opposed to the fixed effects estimator) and earlier lags of endogenous variables as instruments. In this way, the parameters are identified using the within-firm variation in the application of standards and labour productivity over time. Distance to the main buyer, location, legal ownership status and time dummies are treated as exogenous, while standards S_{it} and firm characteristics X_{it} enter estimation as endogeneous. Endogenous variables are instrumented with all available lags (first and second) in the difference equation and with contemporaneous first differences in the levels equation. The validity of all instruments is checked with the Hansen test of over-identification restrictions. The short time series of the panel data (2010-2014)

may limit the extent of variation used to identify parameters and the estimates could be influenced by the exit and entry of firms rather than within-firm variations. Resolving this issue calls for the balanced panel estimates, which I show in addition to the results of the unbalanced panel estimation.

5.1. Descriptive statistics

Table 2 shows the average performance at the firm level by certification of international standards using data from all years. Firms applying standards show two times higher labour productivity levels than non-certified firms. These firms also tend to be larger (employ more full-time workforce) and to have more capital on average. Figure 4 shows that labour productivity increases with firm size, capital and application of standards. This is explored more rigorously in the estimations that follow.

[Table 2 around here]

Returning to Table 2, it is visible that certified firms are also more likely to be younger, to produce intermediate goods and to export. Non-certified firms tend to sell locally with the distance to the main buyer being only 19 km. Firms with standards tend to have more educated owners and a larger proportion of professionals in total work force.

[Figure 4 around here]

6. Results

Table 3 shows the estimates of the impact of international standards on labour productivity among the Vietnamese SMEs from the food sector, where labour productivity is measured by real value added per employee (in 1,000 VND). Equation (1) is estimated using OLS, a firm fixed-effects estimator and the difference GMM estimator. All models include location, legal ownership and time dummies. Column (1) shows the pooled OLS estimates on a balanced panel with location, legal ownership and time fixed effects in addition to the variables reported. The coefficient points to a significant relationship between application of standards and labour productivity.

Column (2) is a counterpart to column (1) with added firm-specific fixed effects which control for all firm-specific time invariant heterogeneity, while column (3) uses the full sample available (unbalanced panel) in

the same type of estimation. As expected, the size of the coefficient drops slightly compared to column (1). The fixed effects estimations show that firms who have adopted private standards enjoy around 20% higher labour productivity than firms that have not adopted such standards. Column (4) contains additional control variables, which increase slightly the magnitude of the coefficient. Columns (5) and (6) show the estimation results for the sample restricted to 2010 and 2012 data as the structure of the questionnaire changed somewhat in 2015. Restricting the sample in such way results in sizeable increase of the coefficient that measures the impact of standards. This shows that the change in the questionnaire structure is not a serious threat to estimation validity.

[Table 3 around here]

Finally, columns (7) and (8) show the results of GMM estimation on unbalanced and balanced panel, employed to address identification challenges in inferring a causal relationship between standards and labour productivity. We find the magnitude of the effect of standards on labour productivity increases by a notably large amount, suggesting a downward bias in the OLS estimates. The magnitude of the coefficient suggests that firms that apply standards increase their labour productivity by 37%. As shown in the lower part of the table, Hansen's test for the validity of the instruments is satisfied.

The results of this study complement earlier findings of the positive impact of standards on work conditions (Colen et al., 2012; Levine and Toffel, 2010; Schuster and Maertens, 2016) and total factor productivity (Goedhuys and Sleuwaegen, 2013). They also support earlier result of the positive impact of environmental standard ISO 14001 on labour productivity among the French firms (Delmas and Pekovic, 2013). Using cross-sectional data, Delmas and Pekovic (2013) have found that the adoption of environmental standards (ISO 14001) is associated with 16-21% increase of labour productivity above the average. The GMM results from present study are quantitatively larger, but the OLS and FE results are in the same order of magnitude. The downward bias of the OLS estimation is likely coming from unobservable characteristics that are negatively correlated with the covariates. The unobserved characteristics which lower the probability of applying standards lead to better labour productivity, indicating perhaps that firms with weaker managerial capabilities are more likely to seek to improve performance through standards, while more capable firms may not need standards for this purpose. This may point to a trade-off between the investment in private standards and labour productivity for financially constrained firms.

Looking at the control variables, the positive and significant coefficient on the firm size in the OLS estimation disappears after controlling for unobserved heterogeneity. The value of assets is, however, consistently positive in all estimations, implying higher labour productivity among more technology-endowed firms. The relationship between firm age and worker wages is negative, but imprecisely determined in all estimations apart from OLS. The relationship between producing final goods and labour productivity is negative, implying better outcomes for firms who produce intermediate goods. The estimates do not show consistent evidence for the returns on export, professional workforce and owner education.

That private standards affect firm operational practices is established in Table A1 in the Appendix, which shows the results of a falsification exercise on the relationship between labour productivity and buyers' requests for certifying standards. Table A1 in the Appendix shows that buyers' requests for certification are not significantly related to labour productivity when firm, location, legal ownership and time effects are accounted for. This implies that standards indeed introduce meaningful operational changes, which, as hypothesised, lead to better operating procedures, management practices, production processes and finally, improved labour productivity.

Table 4 shows the impact of standards on the sub-sample of formally registered firms. The results are very close in significance and magnitude to the estimation in Table 3. As mentioned in Section 3, some firms apply more than one standard, which could affect the precision of estimates if multiple standards bring synergic benefits. To address this challenge, I restrict the sample to formal firms who apply only one standard. Table 5 shows that the results remain very close in significance and magnitude to the original estimation. As it may take time for productivity to reach its long-run level whenever factors of production are changed, I re-estimate Eq. 1 in a dynamic form by introducing lagged dependent variable on the right-hand side. The results shown in Table A2 in the Appendix generally confirm the finding that standards increase labour productivity.

[Table 4 around here]

[Table 5 around here]

The final part of the analysis estimates whether the impact of standards on labour productivity depends on the intensity of labour use. Labour intensity is measured as annual real costs of labour. Table 6 shows that

the impact of standards on labour productivity goes through labour intensity and that the direction of the relationship is convex, with a negative first interaction and a positive interaction on the quadratic term. This indicates that the benefits from standards accrue to firms with labour intensity above a certain threshold. As an illustration, Figure 5 shows how average labour costs and productivity differ by application of standards, comparing the firms with low and high labour productivity and firms with low and high labour intensity, respectively. Panel (a) shows that more productive firms applying standards also have higher labour compensation rates, measured as average real labour costs. Panel (b) shows that more labour intensive firms applying standards also have higher labour productivity.⁴

[Table 6 around here]

[Figure 5 around here]

7. Conclusion

Private standards in developing countries are mostly studied with respect to financial performance and access to export markets, leaving limited little evidence about the impact on employee outcomes, especially on labour productivity. This issue is highly relevant, especially in light of the recent reports of stagnating labour productivity in developing countries (e.g., see Rodrik, 2011).

Earlier research has been ambiguous about the direction of the relationship between private standards and labour productivity. This paper presents evidence that private standards have a positive effect on labour productivity among the SMEs from the food sector in Vietnam, showing that standards can contribute to more than market access and profits. This finding supports earlier research on the relationship between private standards and work conditions (Colen et al., 2012; Delmas and Pekovic, 2013; Levine and Toffel, 2010; Schuster and Maertens, 2016).

It has been argued in earlier research that productivity would increase with the application of standards only if the labour is adequately compensated. The paper shows indeed that productivity gains from standards depend on the level of labour compensation, but this relationship is not linear. Positive productivity effects from standards are observable for highly labour-intensive firms, pointing to the

⁴ Note that the estimation in Table 6 includes a continuous measure of labour costs, while Figure 5 separates firms into groups with below and above median labour costs per employee and below and above median labour productivity.

presence of a threshold level of labour compensation necessary for productivity gains. Firms with low levels of labour intensity are not likely to improve labour productivity by applying private standards. This implies that employee wage increase due to standards is a likely mechanism for further labour productivity gains. This is an important finding in the context of weak institutional environments of developing countries, where substitutes for official regulation in the form of private standards appear to be able to generate additional benefits for SMEs. Indeed, Goedhuys and Sleuwaegen (2013) found that certification increases productivity especially in countries with weak institutional framework.

The results are reliable as they are based on the three rounds of panel data, which allow controlling for confounding effects, such as self-selection and unobserved firm-specific characteristics that may or may not change over time. Time-invariant confounding factors are addressed with firm-specific fixed effects, the influence of region-specific characteristics with location fixed effects and general trends that affect all firms with time dummies. GMM estimation is applied to address the influence of time-varying firm-specific factors. Finally, the results are robust to a number of specification changes: placebo exercise, inclusion of lagged dependent variable, restriction of the sample to only formal firms or to firms who apply only one private standard.

The results could be extended in a couple of ways. First, the application of standards is not very common among the SMEs from the food sector in Vietnam. Moreover, the range of standards applied is quite limited compared to other countries. This has prevented measuring the benefits of different types of standards at the intensive margin. Future work could thus estimate the effect of different types of standards on labour productivity. Second, future work could perhaps focus in greater detail on labour productivity mechanisms, some of which could be due to management or labour force skills.

Tables

Table 1: Summary statistics

Variable	Description	2010		2012		2014	
		Mean	SD	Mean	SD	Mean	SD
Standards	Proportion of firms applying internationally recognised standards (%)	4.58	(20.92)	4.91	(21.63)	3.22	(17.67)
Labour productivity	Real value added per worker (1,000 VND)	19.85	(30.93)	50.53	(51.43)	60.68	(226.56)
Firm size	Total full-time regular labour force	8.95	(20.24)	8.71	(20.18)	8.00	(18.87)
Assets	Real value of total assets (1,000 VND)	1,137	(3,468)	2,619	(9,265)	1,912	(6,491)
Age of the firm	Number of years since the firm has been established	19.25	(10.09)	19.23	(10.55)	18.37	(10.88)
Final goods share	Proportion of output used for final consumption (%)	38.88	(37.25)	48.30	(37.99)	52.17	(38.25)
Distance	Distance to the main buyer in km	23.28	(73.30)	19.22	(48.76)	24.16	(65.23)
Export	Firm sells to foreign countries (%)	3.84	(19.23)	2.89	(16.77)	2.84	(16.61)
Owner has higher education	Owner has completed secondary education (%)	44.46	(49.73)	52.60	(49.97)	55.80	(49.69)
Professionals share	Proportion of professional workers in a firm (%)	1.83	(5.68)	1.36	(4.29)	0.94	(3.48)
Formally registered firms	Proportion of formally registered firms (%)	56.87	(49.56)	55.35	(49.75)	82.99	(37.60)
<i>Legal ownership form</i>							
Household establishment	Proportion of firms listed as household establishment (%)	85.52	(35.21)	84.97	(35.76)	86.08	(34.64)
Private/sole proprietorship	Proportion of firms listed as private or sole owner establishment (%)	4.58	(20.92)	4.05	(19.72)	2.96	(16.97)
Partnership/ Collective/ Cooperative	Proportion of firms listed as partnership, collective or cooperative (%)	0.30	(5.43)	0.87	(9.28)	0.52	(7.17)
Limited liability company	Proportion of firms listed as limited liability company (%)	8.27	(27.57)	8.24	(27.51)	8.63	(28.10)
Joint stock company	Proportion of firms listed as joint stock company (%)	1.33	(11.46)	1.88	(13.59)	1.80	(13.32)
Observations		677		692		776	

Note: Average 2010 exchange rate: 1 USD = 19,128 VND.

Table 2: Differences between firms by implementation of standards, 2010-2014.

Variable	All	Standards	No standards	Difference	t-value
Labour productivity (ln)	44.52 (141.44)	100.68 (121.20)	42.06 (141.78)	58.62	3.86***
Firm size (ln)	8.53 (19.73)	50.34 (47.47)	6.70 (15.12)	43.64	22.92***
Assets (ln)	1,895 (6,858)	14,405 (22,641)	1,348 (4,442)	13,058	19.12***
Age of the firm (years)	18.92 (10.54)	16.71 (11.02)	19.02 (10.51)	-2.31	-2.04**
Final goods share (%)	46.73 (38.24)	25.94 (34.09)	47.64 (38.16)	-21.69	-5.30***
Distance (km)	22.29 (63.27)	87.85 (156.78)	19.42 (54.00)	68.43	10.29***
Export (%)	3.17 (17.52)	44.44 (49.97)	1.36 (11.60)	43.08	26.23***
Owner has higher education (%)	51.19 (50.00)	94.44 (23.03)	49.29 (50.01)	45.15	8.52***
Professionals (%)	1.35 (4.54)	8.32 (7.83)	1.05 (4.08)	7.27	15.69***

Note: Average 2010 exchange rate: 1 USD = 19,128 VND. Standard deviation in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01. Number of observations: 2,145.

Table 3: Impact of standards on labour productivity. Dependent variable: value added per worker (ln).

	OLS balanced	FE balanced	FE unbalanced	FE balanced	FE unbalanced, 2010-2012	FE balanced, 2010-2012	GMM unbalanced	GMM balanced
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standards	0.216** (0.097)	0.200** (0.078)	0.177** (0.075)	0.186** (0.081)	0.302** (0.120)	0.289** (0.133)	0.343** (0.173)	0.318** (0.132)
Firm size (ln)	0.087** (0.037)	-0.061 (0.057)	-0.036 (0.049)	-0.062 (0.057)	-0.084 (0.063)	-0.066 (0.069)	-0.011 (0.131)	-0.043 (0.112)
Assets (ln)	0.162*** (0.019)	0.145*** (0.025)	0.153*** (0.022)	0.147*** (0.025)	0.113*** (0.030)	0.116*** (0.032)	0.194*** (0.047)	0.178*** (0.041)
Age of the firm	-0.005*** (0.002)	0.003 (0.006)	0.005 (0.006)	0.002 (0.006)	0.008 (0.006)	0.005 (0.007)	-0.006 (0.021)	-0.008 (0.018)
Final goods share	-0.002*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.002** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.002* (0.001)	-0.001 (0.001)
Distance (ln)	0.014 (0.013)	0.009 (0.014)	0.008 (0.012)	0.009 (0.014)	0.009 (0.017)	0.013 (0.018)	0.013 (0.013)	0.014 (0.014)
Export				0.125 (0.138)	-0.218 (0.182)	-0.203 (0.188)	0.254 (0.338)	0.117 (0.339)
Owner has higher education				0.001** (0.001)	0.002*** (0.001)	0.002*** (0.001)	-0.000 (0.001)	-0.000 (0.001)
Professionals				0.002 (0.006)	0.011 (0.007)	0.010 (0.008)	0.003 (0.012)	-0.009 (0.011)
Year dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Legal form dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.824*** (0.135)	1.853*** (0.200)	1.690*** (0.173)	1.807*** (0.198)	1.889*** (0.240)	1.916*** (0.259)		
Number of obs.	1,425	1,425	2,144	1,425	1,369	950	1,146	950
Number of firms		475	988	475	829	475	668	475
R ²	0.61	0.64	0.63	0.64	0.75	0.75		
Hansen test statistics							4.55	5.77
Hansen test p-value							0.47	0.22

Note: Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 4: Impact of standards on labour productivity: estimation on the subsample of formal firms.
Dependent variable: value added per worker (ln).

	OLS balanced	FE balanced	FE unbalanced	FE balanced	FE unbalanced, 2010-2012	FE balanced, 2010-2012	GMM unbalanced	GMM balanced
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standards	0.219** (0.098)	0.195** (0.076)	0.204*** (0.078)	0.201** (0.080)	0.299** (0.120)	0.275** (0.129)	0.332** (0.158)	0.380*** (0.129)
Firm size (ln)	0.066 (0.043)	0.013 (0.060)	-0.035 (0.069)	-0.034 (0.069)	-0.056 (0.075)	-0.057 (0.082)	-0.053 (0.126)	-0.041 (0.111)
Assets (ln)	0.164*** (0.023)	0.152*** (0.025)	0.155*** (0.028)	0.155*** (0.029)	0.118*** (0.040)	0.121*** (0.043)	0.187*** (0.051)	0.169*** (0.045)
Age of the firm	-0.005** (0.002)	-0.004 (0.009)	-0.008 (0.010)	-0.008 (0.010)	0.001 (0.010)	-0.005 (0.010)	0.025 (0.028)	0.013 (0.023)
Final goods share	-0.002** (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)	-0.001 (0.001)
Distance (ln)	0.023 (0.016)	0.020 (0.016)	0.022 (0.018)	0.021 (0.018)	0.026 (0.021)	0.028 (0.023)	0.022 (0.014)	0.019 (0.015)
Export				0.068 (0.157)	-0.210 (0.181)	-0.172 (0.185)	0.258 (0.331)	0.135 (0.355)
Owner has higher education				0.001 (0.001)	0.002** (0.001)	0.002* (0.001)	0.000 (0.001)	-0.000 (0.001)
Professionals				0.002 (0.006)	0.010 (0.007)	0.009 (0.008)	0.004 (0.012)	-0.009 (0.011)
Year dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Legal form dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.828*** (0.156)	1.806*** (0.238)	1.979*** (0.279)	1.924*** (0.278)	2.091*** (0.395)	2.208*** (0.432)		
Number of obs.	983	1,411	983	983	768	525	885	724
Number of firms		783	468	468	474	267	626	468
R ²	0.53	0.67	0.68	0.68	0.77	0.77		
Hansen test statistics							4.34	5.22
Hansen test p-value							0.50	0.27

Note: Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 5: Impact of standards on labour productivity: estimation on the subsample of formal firms that apply only one standard. Dependent variable: value added per worker (ln).

	OLS balanced	FE balanced	FE unbalanced	FE balanced	FE unbalanced, 2010-2012	FE balanced, 2010-2012	GMM unbalanced	GMM balanced
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standards	0.217** (0.108)	0.197** (0.086)	0.205** (0.082)	0.203** (0.085)	0.299** (0.120)	0.275** (0.129)	0.455** (0.198)	0.374*** (0.142)
Firm size (ln)	0.069 (0.044)	0.018 (0.061)	-0.030 (0.072)	-0.029 (0.072)	-0.056 (0.075)	-0.057 (0.082)	-0.080 (0.130)	-0.054 (0.128)
Assets (ln)	0.165*** (0.023)	0.141*** (0.027)	0.143*** (0.030)	0.143*** (0.030)	0.118*** (0.040)	0.121*** (0.043)	0.163*** (0.055)	0.164*** (0.049)
Age of the firm	-0.005** (0.002)	-0.002 (0.008)	-0.006 (0.008)	-0.006 (0.008)	0.001 (0.010)	-0.005 (0.010)	0.027 (0.033)	0.019 (0.028)
Final goods share	-0.002** (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.002* (0.001)	-0.002* (0.001)	-0.001 (0.001)	-0.001 (0.001)
Distance (ln)	0.023 (0.016)	0.022 (0.017)	0.024 (0.019)	0.023 (0.019)	0.026 (0.021)	0.028 (0.023)	0.017 (0.015)	0.020 (0.016)
Export				0.028 (0.164)	-0.210 (0.181)	-0.172 (0.185)	0.108 (0.316)	0.008 (0.344)
Owner has higher education				0.001 (0.001)	0.002** (0.001)	0.002* (0.001)	0.001 (0.001)	-0.000 (0.001)
Professionals				0.001 (0.006)	0.010 (0.007)	0.009 (0.008)	-0.013 (0.013)	-0.017* (0.010)
Year dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Legal form dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.817*** (0.154)	1.811*** (0.247)	1.984*** (0.287)	1.937*** (0.287)	2.091*** (0.395)	2.208*** (0.432)		
Number of obs.	966	1386	966	966	768	525	863	707
Number of firms		779	466	466	474	267	620	466
R ²	0.52	0.68	0.69	0.69	0.77	0.77		
Hansen test statistics							7.09	7.06
Hansen test p-value							0.21	0.13

Note: Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Impact of standards on labour productivity depending on the real labour costs per employee (ln).
Dependent variable: real labour costs per employee (ln).

	FE balanced	FE unbalanced	FE balanced	FE unbalanced, 2010-2012	FE balanced, 2010-2012	GMM unbalanced	GMM balanced
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Standards	0.215*** (0.083)	0.218** (0.092)	0.203** (0.096)	0.357** (0.155)	0.393** (0.178)	0.411* (0.218)	0.172 (0.180)
Standards*Labour costs interaction	-0.000 (0.000)	-0.000* (0.000)	-0.000* (0.000)	-0.001** (0.000)	-0.001** (0.000)	-0.000 (0.000)	-0.000 (0.000)
Standards*Labour costs sq. interaction	0.000* (0.000)	0.000 (0.000)	0.000* (0.000)	0.000* (0.000)	0.000** (0.000)	0.000 (0.000)	0.000 (0.000)
Labour costs	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
Labour costs sq.	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)
Constant	1.782*** (0.223)	2.004*** (0.253)	1.985*** (0.257)	1.979*** (0.322)	2.076*** (0.360)		
Number of obs.	1,476	999	999	1,075	750	614	501
Number of firms	838	475	475	766	475	382	275
R ²	0.70	0.71	0.72	0.79	0.79		
Hansen test statistics						37.33	25.17
Hansen test p-value						0.00	0.01

Note: Each model includes following covariates: firm size (ln), real value of assets (ln), firm age, final goods, distance to the main buyer (ln), export status, owner having higher education, professional workforce, time, legal ownership form and location fixed effects. Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Figures

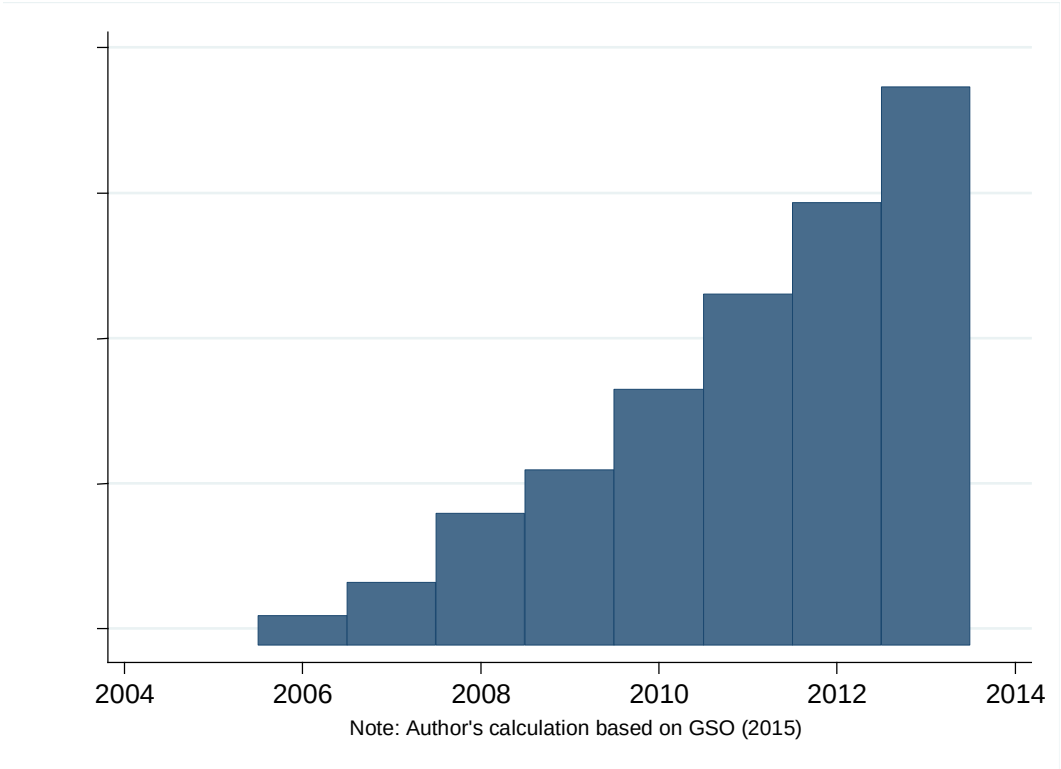


Figure 1: Gross output of food manufacturing (at current prices).

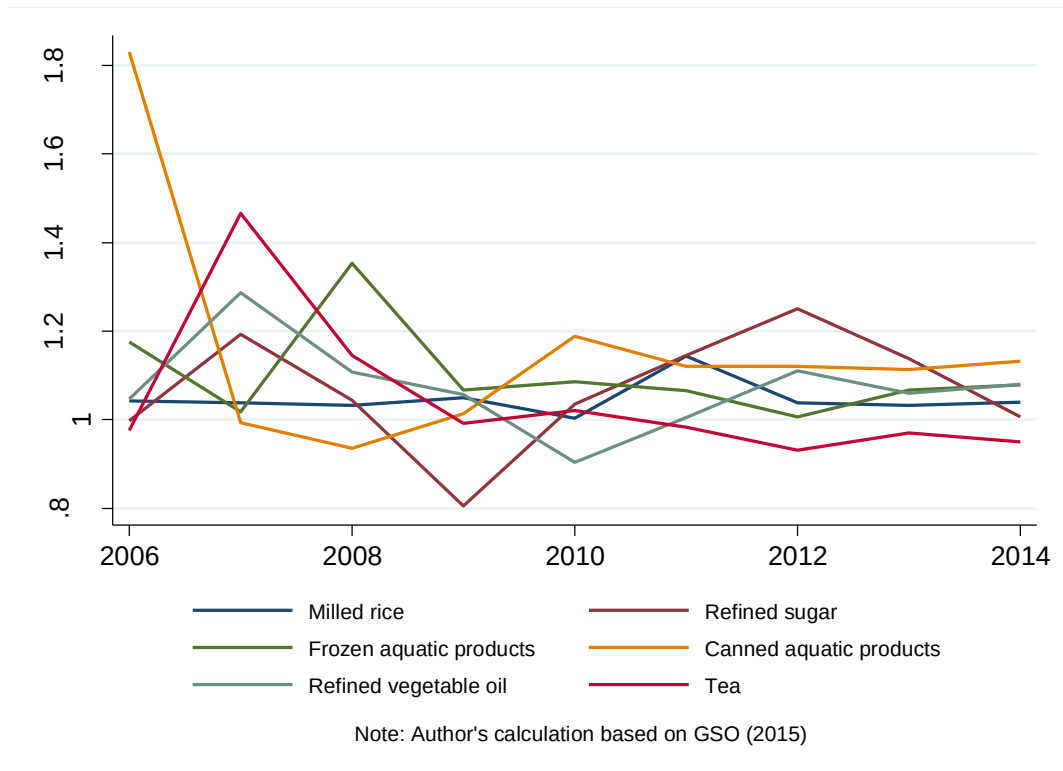


Figure 2: Main products in the Vietnamese food manufacturing sector.

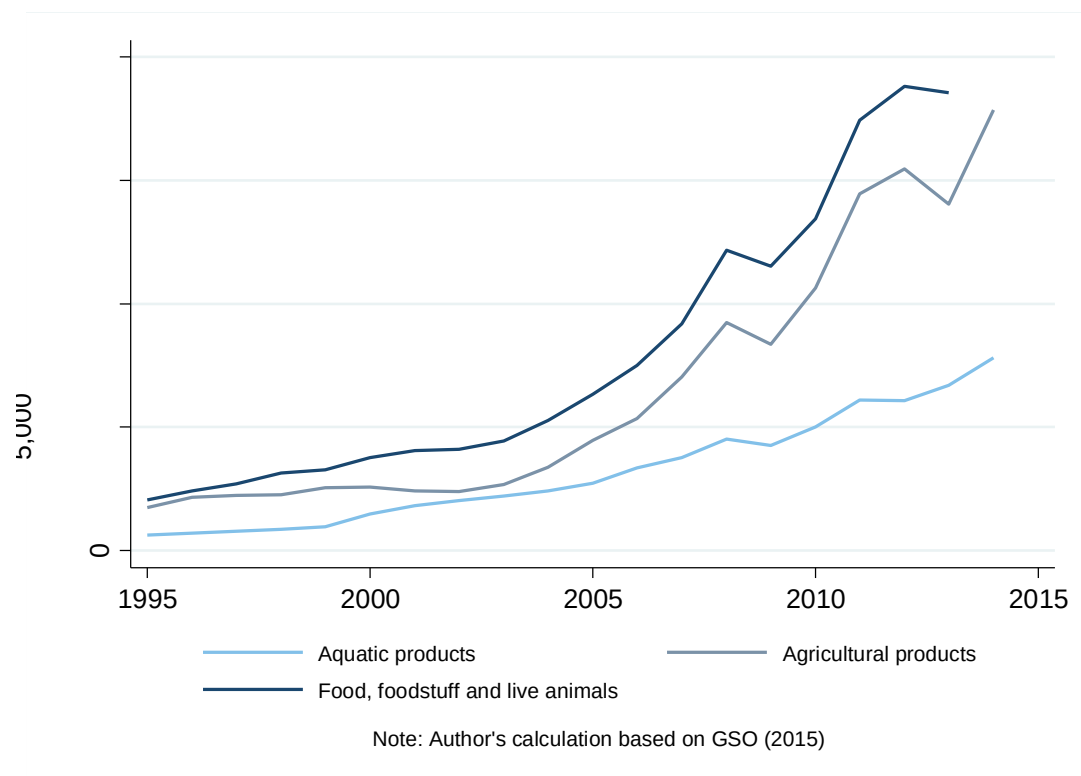


Figure 3: Export of food from Vietnam.

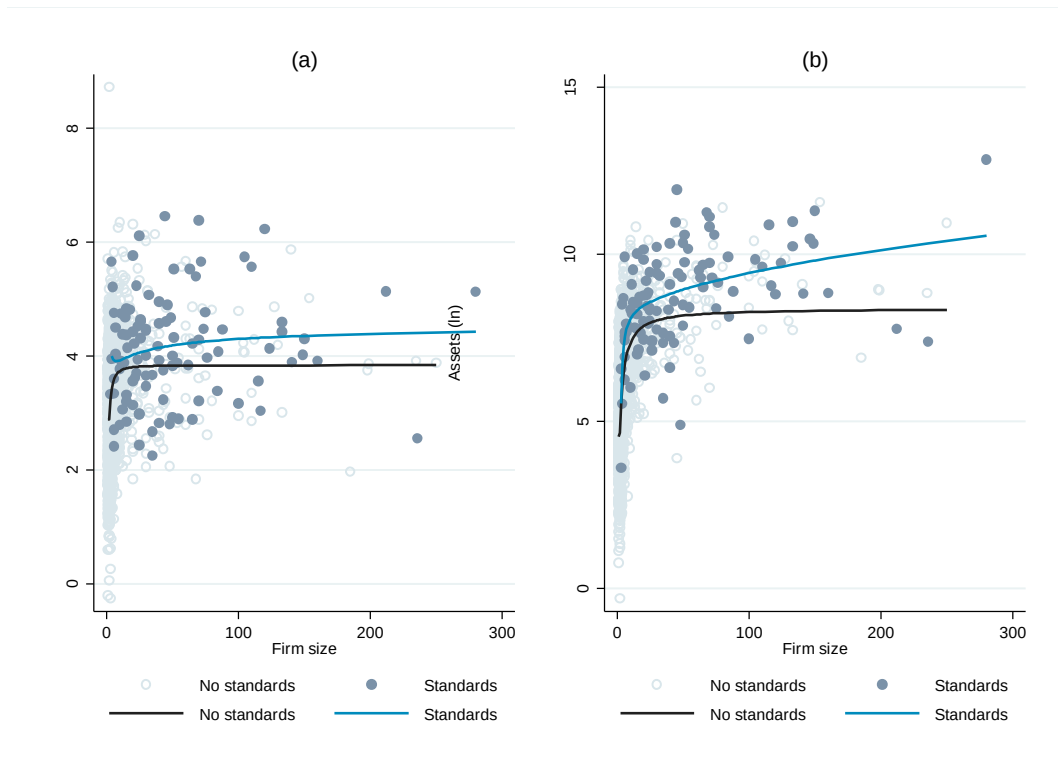


Figure 4: Labour productivity and real value of assets by application of standards.

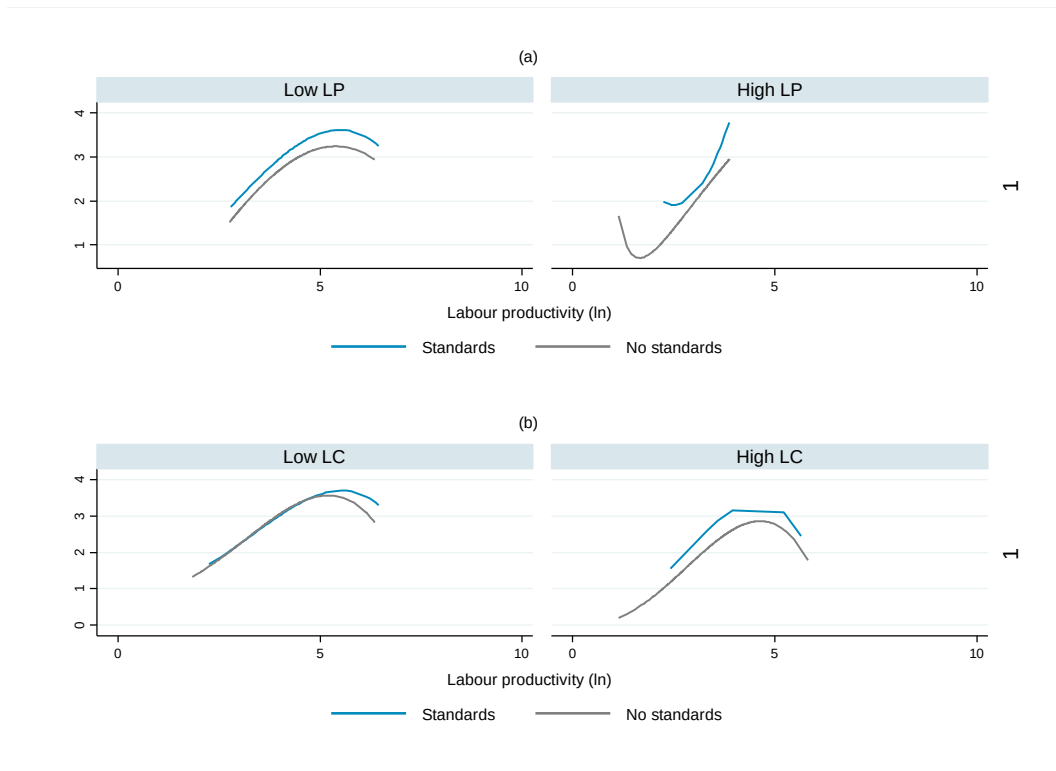


Figure 5: Average labour costs by application of standards among firms with low and high labour productivity and intensity, respectively.

Note: Panel (a) separates firms based on labour productivity (LP) measured as real value added per employee into firms with low and high productivity where 'Low LP' comprises firms with LP below median and 'High LP' groups firms with LP above median. Panel (b) separates firms based on labour costs (LC), measured as real value of labour at the end of previous accounting year. 'Low LC' are firms with LC below median and 'High LC' are firms with LC higher than median.

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Appendix

Table A1: Placebo test: Impact of requesting private standards on labour productivity. Dependent variable: value added per worker (ln).

	OLS balanced	FE balanced	FE unbalanced	FE balanced	FE unbalanced, 2010-2012	FE balanced, 2010-2012	GMM unbalanced	GMM balanced
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Buyers' request for standards	0.083 (0.075)	-0.010 (0.088)	-0.054 (0.080)	-0.021 (0.091)	-0.135 (0.265)	-0.076 (0.293)	-0.015 (0.155)	-0.020 (0.179)
Firm size (ln)	0.089** (0.037)	-0.063 (0.057)	-0.038 (0.049)	-0.065 (0.057)	-0.087 (0.065)	-0.067 (0.070)	-0.088 (0.111)	-0.068 (0.116)
Assets (ln)	0.162*** (0.019)	0.147*** (0.025)	0.156*** (0.022)	0.150*** (0.025)	0.120*** (0.029)	0.122*** (0.031)	0.179*** (0.046)	0.181*** (0.042)
Age of the firm	- 0.005*** (0.002)	0.003 (0.006)	0.005 (0.006)	0.002 (0.006)	0.009 (0.006)	0.005 (0.007)	0.000 (0.020)	-0.007 (0.018)
Final goods share	- 0.002*** (0.001)	-0.002** (0.001)	-0.001** (0.001)	-0.002** (0.001)	-0.003*** (0.001)	-0.003*** (0.001)	-0.002* (0.001)	-0.001 (0.001)
Distance (ln)	0.014 (0.013)	0.009 (0.014)	0.007 (0.012)	0.009 (0.014)	0.008 (0.017)	0.012 (0.018)	0.012 (0.013)	0.014 (0.014)
Export				0.193 (0.127)	0.185 (0.248)	0.139 (0.261)	0.357 (0.303)	0.217 (0.354)
Owner has higher education				0.001** (0.001)	0.002*** (0.001)	0.002*** (0.001)	-0.000 (0.001)	-0.000 (0.001)
Professionals				0.001 (0.006)	0.010 (0.007)	0.008 (0.008)	0.004 (0.011)	-0.012 (0.010)
Year dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Location dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Legal form dummies		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Constant	1.882*** (0.137)	1.850*** (0.196)	1.681*** (0.169)	1.800*** (0.194)	1.841*** (0.227)	1.878*** (0.244)		
Number of obs.	1425	1425	2144	1425	1369	950	1158	950
Number of firms		475	988	475	829	475	675	475
R ²	0.61	0.64	0.63	0.64	0.75	0.75		
Hansen test statistics							5.49	7.40
Hansen test p- value							0.48	0.12

Note: Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.

Table A2: Dynamic estimation of the impact of standards on labour productivity. Dependent variable: value added per worker (ln).

	OLS balanced (1)	FE balanced (2)	FE unbalanced (3)	FE balanced (4)	GMM balanced (5)	GMM unbalanced (6)
Standards	0.044 (0.114)	0.300* (0.180)	0.309* (0.162)	0.258 (0.192)	0.613 (0.381)	0.899** (0.413)
Lag labour productivity (ln)	0.348*** (0.034)	-0.312*** (0.046)	-0.317*** (0.046)	-0.313*** (0.045)	0.201** (0.089)	0.162* (0.089)
Firm size (ln)	0.033 (0.037)	-0.135* (0.077)	-0.134* (0.076)	-0.141* (0.077)	-0.177 (0.192)	-0.213 (0.185)
Assets (ln)	0.114*** (0.021)	0.152*** (0.033)	0.156*** (0.032)	0.153*** (0.033)	0.126 (0.081)	0.096 (0.087)
Age of the firm	-0.003 (0.002)	-0.038 (0.035)	-0.038 (0.035)	-0.040 (0.036)	0.120 (0.177)	0.136 (0.201)
Final goods share	-0.002** (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.000 (0.001)	-0.002 (0.002)	-0.002 (0.002)
Distance (ln)	0.005 (0.015)	0.010 (0.017)	0.010 (0.017)	0.009 (0.017)	0.010 (0.021)	0.010 (0.020)
Export				0.177 (0.171)	0.030 (0.350)	-0.156 (0.349)
Owner has higher education				-0.000 (0.001)	0.000 (0.002)	-0.000 (0.002)
Professionals				-0.003 (0.008)	0.017 (0.023)	0.023 (0.022)
Year dummies		Yes	Yes	Yes	Yes	Yes
Location dummies		Yes	Yes	Yes	Yes	Yes
Legal form dummies		Yes	Yes	Yes	Yes	Yes
Constant	2.225*** (0.169)	4.474*** (0.818)	4.447*** (0.797)	4.533*** (0.836)		
Number of obs.	950	950	1158	950	475	492
Number of firms		475	675	475	475	492
R ²	0.46	0.16	0.17	0.16		
Hansen test statistics					2.15	1.13
Hansen test p-value					.	.

Note: Robust standard errors clustered at the firm level are in parentheses. Significance levels: * p<0.10, ** p<0.05, *** p<0.01.